

Chapter 19

Open-Economy Macroeconomics

: The Balance of Payments and Exchange Rates

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- ① Equilibrium Output in an Open Economy
- ② Exchange Rates
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The International Sector and Planned Aggregate Expenditure

- Planned aggregate expenditure in an open economy:

$$AE \equiv C + I + G + \underbrace{EX - IM}_{\text{Net Export}}.$$

- Exports represent the demand for domestic products by the rest of the world.
- Note that imports are not a part of domestic output because they are produced outside the home country.
- When we calculate households' total consumption spending, firms' total investment spending, and total government spending, imports are included.
- Therefore, to calculate domestic output correctly, we must subtract the parts of consumption, investment, and government spending that constitute imports.

Determining the Level of Imports

- What determines the level of imports and exports in a country?
- Clearly the level of imports is a function of current income Y .
 - When U.S. income increases, U.S. citizens buy more of everything, including Japanese cars and Korean smartphones.

$$IM = mY.$$

where m is the marginal propensity to import (MPM).

- For now, we will assume that exports are given.

Solving for Equilibrium

- Imposing the equilibrium condition,

$$\begin{aligned}Y &= AE \\&= C + I + G + EX - IM \\&= a + b(Y - T) + I + G + EX - mY.\end{aligned}$$

$$\rightarrow Y^* = \frac{a + I + G - bT + EX}{1 - (b - m)}.$$

- $(b - m)$ is referred to as MPC *domestically* produced goods.
- Multiplier effects:
 - Which variables are exogenous in this case?
 - Comparison between closed-economy multipliers and open-economy multipliers

A Realistic Description of Import

- In reality, the amount of spending on imports also depends on factors other than income and exports are not fixed.
- The Determinants of Imports
 - Note that imports are the *consumption* of goods produced in foreign countries.
 - Therefore, the same factors that affect households' consumption behavior and firms' investment behavior are likely to affect the demand for imports.
 - After-tax real wage, after-tax nonlabor income, interest rate, and so on.
 - Relative prices of domestically produced and foreign-produced goods.

The Determinants of Exports

- Remember that countries are interdependent with an open economy.
- The foreign demand for U.S. exports is identical to the foreign countries' imports from the United States.
- Thus, the demand for U.S. exports depends on economic activity in the rest of the world as well as on the prices of U.S. goods relative to the prices of rest-of-the-world goods.

The Trade Feedback Effect

- The tendency for an increase in the economic activity of one country to lead to a worldwide increase in economic activity, which then feeds back to that country.
- Suppose the United States finds its exports increasing, perhaps because the world suddenly decides that it prefers U.S. computers to other computers.
- Exports of U.S. $\uparrow \rightarrow$ Income of U.S. $\uparrow \rightarrow$ Imports of U.S. $\uparrow \rightarrow$ Exports of the rest of the world to U.S. $\uparrow \rightarrow \dots$

The Price Feedback Effect

- Note that an increase in domestic prices increases the prices of its exports.
 - It is also true that when the prices of a country's imports increase, the prices of domestic goods may increase in response.
- The process by which a domestic price increase in one country can "feed back" on itself through export and import prices.
 - An increase in the price level in one country can drive up prices in other countries.
 - This in turn further increases the price level in the first country.

The Market for Foreign Exchange

- Assume exchange rates follow a floating rate system and there are just two countries: the United States and Great Britain (foreign country).

TABLE 19.2 Some Buyers and Sellers in International Exchange Markets: United States and Great Britain

The Demand for Pounds (Supply of Dollars)

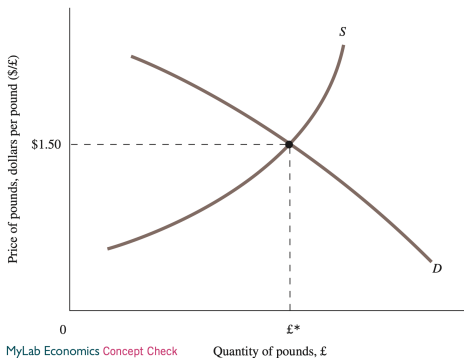
1. Firms, households, or governments that import British goods into the United States or want to buy British-made goods and services
2. U.S. citizens traveling in Great Britain
3. Holders of dollars who want to buy British stocks, bonds, or other financial instruments
4. U.S. companies that want to invest in Great Britain
5. Speculators who anticipate a decline in the value of the dollar relative to the pound

The Supply of Pounds (Demand for Dollars)

1. Firms, households, or governments that import U.S. goods into Great Britain or want to buy U.S.-made goods and services
 2. British citizens traveling in the United States
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 4. British companies that want to invest in the United States
 5. Speculators who anticipate a rise in the value of the dollar relative to the pound
-

The Supply of and Demand for Pounds (Foreign Exchange)

- The key to understanding the supply and demand curves represented here is to recognize that the price on the vertical axis is the price of one currency relative to a second.



Change of Exchange Rate

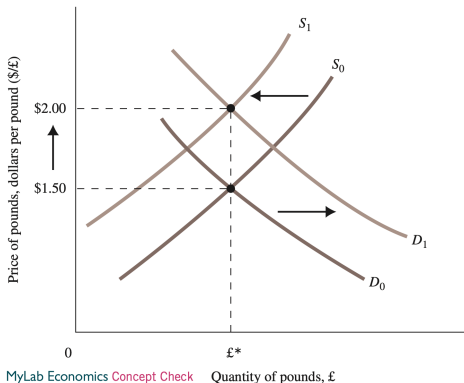
- Appreciation of a currency: The rise in the price of one currency relative to another.
 - Price of exports $\uparrow$, Price of imports $\downarrow \rightarrow$ Net exports $\downarrow$
- Depreciation of a currency: the fall in the price of one currency relative to another.
 - Price of exports $\downarrow$, Price of imports $\uparrow \rightarrow$ Net exports $\uparrow$

Purchasing Power Parity: The Law of One Price

- Law of one price
 - If the transportation costs are small, the price of the same good in different countries should be roughly the same.
 - The price of a good should be the same regardless of where we buy it as price differences across the two countries create arbitrage opportunities.
- Purchasing-power-parity theory
 - A theory of international exchange holding that exchange rates are set so that the price of similar goods in different countries is the same.

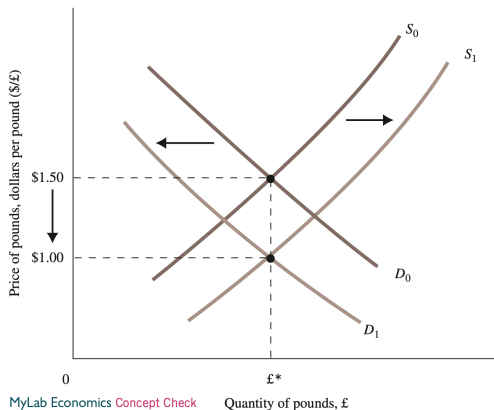
Exchange Rates Respond to Changes in Relative Prices

- An increase in the U.S. price level relative to the price level in Great Britain.



Relative Interest Rates

- If the interest rate is 2 percent in the United States and 3 percent in Great Britain, people with money to lend have an incentive to buy British securities instead of U.S. securities.



The Effects of Exchange Rates on the Economy

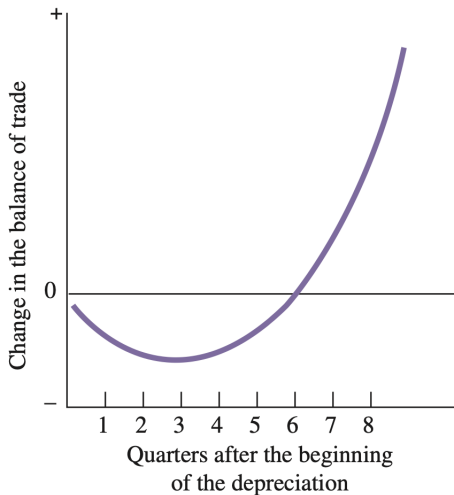
- A depreciation (falls in value) of a country's currency is likely to increase its GDP.
 - Import prices $\uparrow$ and export prices $\downarrow \rightarrow$ net exports $\uparrow$
- J-curve effect
 - Following a currency depreciation, a country's balance of trade may get worse before it gets better. The graph showing this effect is shaped like the letter *J*, hence the name J-curve effect.

Balance of Trade

= dollar price of exports $\times$ quantity of exports

– dollar price of imports $\times$ quantity of imports

The J Curve



Exchange Rate and Prices

- The depreciation of a country's currency tends to increase its price level.
 - Price competitiveness of domestic goods $\uparrow \rightarrow$ exports $\uparrow$
 - Domestic consumers substitute domestic products for the now-more-expensive imports $\rightarrow$ rightward shift of AD curve

Monetary Policy with Flexible Exchange Rates

- Suppose the Fed lowers the interest rate to stimulate the economy.
 - The lower interest rate stimulates planned investment spending and consumption spending (lenders, substitution effect > income effect).
 - A lower interest rate means a lower demand for U.S. securities by foreigners, so the demand for dollars drops.
 - U.S. investment managers will be more likely to buy foreign securities, which are now paying relatively higher interest rates, so the supply of dollars rises.
 - Both events push down the value of the dollar.
- Flexible exchange rates thus help the Fed in its goal to stimulate the economy.
- Flexible exchange rates help to cool down the economy as well (when the Fed raises the interest rate).

Fiscal Policy with Flexible Exchange Rates

- Flexible exchange rates makes the task of the fiscal authorities to achieve its goals more difficult.
- Suppose that the government increases government spending to stimulate the economy.
 - This increases output in the usual way, shift of the AD curve to the right.
 - This usual way means that the interest rate is also higher.
 - The higher interest rate attracts foreign investment and leads to an *appreciation* of the dollar.
 - An appreciation increases imports and decreases exports., which has a negative effect on output.
- Thus, the increase in output is less than it would have been had there been no appreciation.
- Flexible exchange rates also hurt the government if they want to contract the economy to fight inflation.

The Balance of Payments

- The record of a country's transactions in goods, services, and assets with the rest of the world; also the record of a country's sources (supply) and uses (demand) of foreign exchange.
- The balance of payments pertains to flows.
- The balance of payments is divided into two major accounts: the current account and the financial account.

U.S. Balance of Payments, 2017

TABLE 19.1 U.S. Balance of Payments, 2017

Current Account		Billions of dollars
(1)	Goods exports	\$1,550.7
(2)	Goods imports	2361.9
(3)	Exports of services	780.9
(4)	Imports of services	538.1
(5)	Balance of trade: $(1) - (2) + (3) - (4)$	-568.4
(6)	Investment income	926.9
(7)	Investment payments	709.0
(8)	Transfer income	149.7
(9)	Transfer payments	264.5
(10)	Balance on current account: $(5) + (6) - (7) + (8) - (9)$	-466.2
Financial account		
(11)	Net capital transfer receipts	24.8
(12)	Change in net U.S. liabilities	375.5
(13)	Net receipts from financial derivatives	-26.4
(14)	Statistical discrepancy	92.2
(15)	Balance of payments: $(10) + (11) + (12) + (13) + (14)$	0.0

Item (13) is the change in foreign assets in the United States minus the change in U.S. assets abroad. In 2017 this number was positive, which means that there was an increase in net U.S. liabilities.

Source: Bureau of Economic Analysis, March 21, 2018.

The Current Account

- Balance of trade: A country's exports of goods and services minus imports of goods and services.
- Trade deficit: The situation when a country imports more than it exports.
- Invest income and investment payments
 - Earnings and costs from holding foreign assets: dividends, interest, rent, and profits
- Transfer income and transfer payments
 - Relief support, remittances
 - money transfer without expecting anything in return

The Financial Account

- For each transaction recorded in the current account, there is an *offsetting* recorded in the financial account.
 - For example, the \$466.2 billion current account deficit must be paid for, and how it is paid shows up in the financial account.
- Capital receipts, borrowing, receipts from financial derivatives.
- The total measurement error is called the statistical discrepancy.
- The main point to take away is that aside from measurement errors, a current account deficit must be financed by changes in a country's net capital receipts and its net liabilities to the rest of the world.
- The balance of payments is always zero.